Quantum Tunneling in Relativistic Systems

"CERN on one's desk"

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INTRODUCTION

- **Graphene** is a honeycomb-like structure made up of carbon atoms. It is a monolayer of graphite, the material that makes up the tip of pencils.
- Low energy charge carriers exhibit strange behavior in graphene - they behave as if they were **massless relativistic quasiparticles**. This finally allowed people to test counterintuitive relativistic quantum phenomena such as the **Klein paradox [1]** - the anomalous tunneling of electrons with probability 1 when they strike a barrier perpendicularly.
- For these quasiparticles, P-N junctions in graphene behave like lenses with **negative refractive indices**. This allows another physical realization of Veselago lenses, which can be used to focus light/electrons at one point.

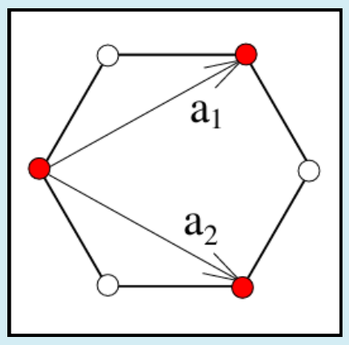
OBJECTIVES

- To study **Dirac's equation** - understand its origin in 3+1 dimensions and 2+1 dimensions and derive Schrödinger's equation from it.
- To study the mapping of graphene's band structure with Dirac's equation near the **Dirac points** and then look at relativistic phenomena in 2D such as Klein Tunneling.
- To understand **catastrophe theory [2]** and apply it to the problem of finding caustics, which is used in the theory of Veselago lenses in graphene.
- To apply all acquired learnt to the problem of **hexagonal altermagnets.**

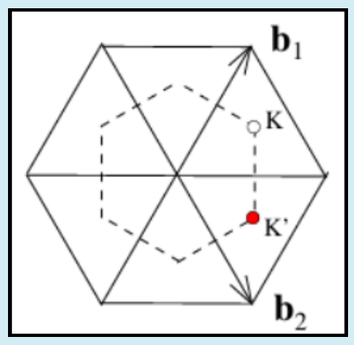
BRIEF OVERVIEW

- Dirac's equation gives a quantum mechanical description of relativistic particles with **spin ½** (unlike the Klein Gordon equation, which describes particles of spin 0).
- Graphene's band structure, derived using the nearest neighbor **tight binding model [3],** is such that on the edges of the first Brillouin zone, the energy dispersion is a linear function of the momentum [Figure 1: (c)].
- This allows a mapping of these low energy electrons to the 2D Dirac equation. The wavefunction is described by **4 components** - 2 because there are two sublattices and 2 for two inequivalent Dirac points [Figure 1: (b)].
- A paradox/consequence of Dirac's equation is that particles can tunnel through a barrier with **perfect transparency**. This result can be obtained by matching wavefunctions at the boundaries of a **rectangular potential barrier.** The computational result of this is shown in Figure 1: (d).
- The theoretical foundations of materials with **negative refractive indices** can be traced back to the work of Veselago in 1968. Light beams get transmitted to the same side of the normal as the incident rays [Figure 1: (e)]. This gives us a method of focusing light beams.
- This same effect can be realized in graphene **[4]** through the means of a PN junction as a result of the fact the **group velocity** and wavevector are in the opposite direction in the P side.
- The quantum mechanical calculations **[5]** require the usage of **Green's functions**, which we use to calculate the wavefunction generated by a poit source (modelled as a delta function), with electrons of different **polarizations**. The symmetric case matches with the classical expectation.

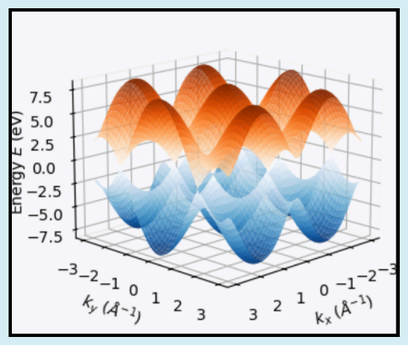
VISUALS AND RESULTS



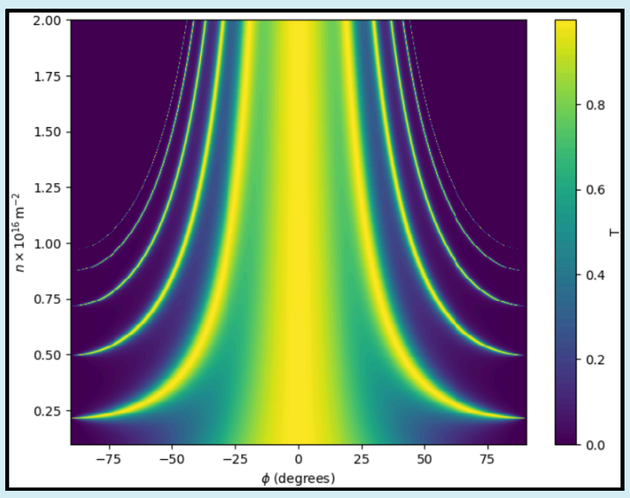
(a)



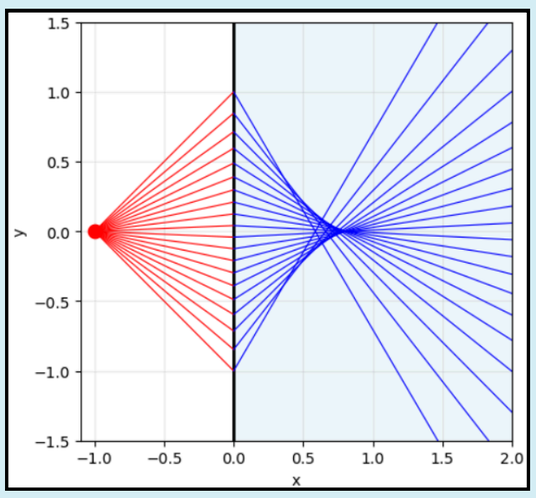
(b)



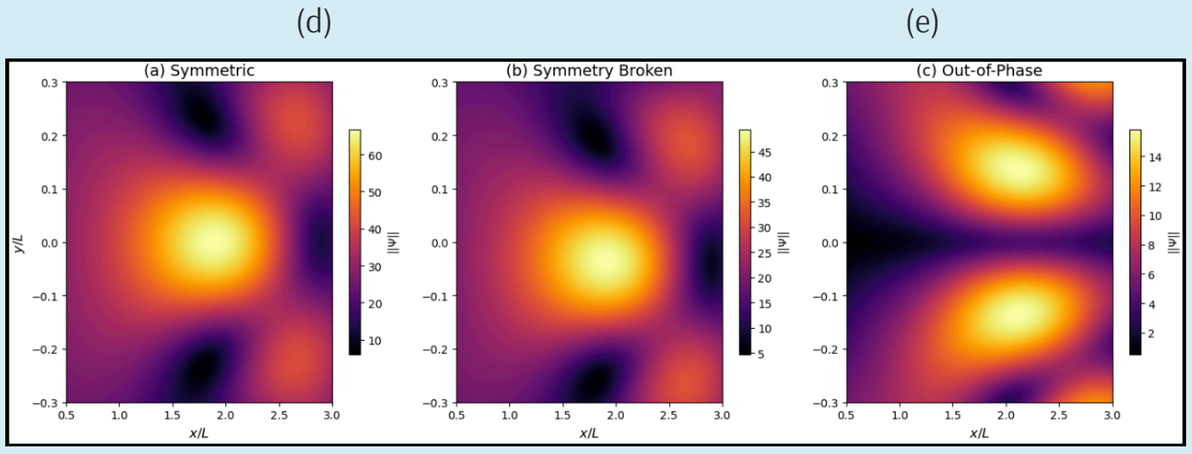
(c)



(d)



(e)



(f)

Figure 1: (a) Graphene lattice in position space. Red dots indicate sublattice A and white dots indicate sublattice B. (b) Graphene lattice in reciprocal space. The dotted hexagon shows the first brillouin zone. (c) Band structure of graphene. Near the Dirac points, the dispersion is linear as can be seen by the cones. (d) Color plot of transmission probability as a function of incidence angle and electron concentration in the n-side. (e) Classical trajectories of electron emanating from a point source incident on a slab of refractive index n = -0.82. The cusp is situated at x = 0.82 units. (f) Color plots of probability density in the p side of the junction obtained through quantum mechanical calculations for 3 cases that differ by the electron psuedospin polarization.

REFERENCES

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[2] M. V. Berry and C. Upstill: Catastrophe Optics: Morphologies of Caustics and Their Diffraction Patterns, in Progress in Optics, edited by E. Wolf (North-Holland, 1980), pp. 257–346.
[3] W. P. Lima, F. R. V. Araújo, D. R. da Costa, S. H. R. Sena, and J. M. Pereira Jr., Tight-binding Model in First and Second Quantization for Band Structure Calculations, Braz. J. Phys. 52, 42 (2022).
[4] V. V. Cheianov, V. I. Fal'ko, and B. L. Altshuler, The Focusing of Electron Flow and a Veselago Lens in Graphene p-n Junctions, Science 315, 1252–1255 (2007).
[5] K. J. A. Reijnders and M. I. Katsnelson, Symmetry breaking and (pseudo)spin polarization in Veselago lenses for massless Dirac fermions, Phys. Rev. B 95, 115310 (2017).

FUTURE PROSPECTS

- Solve the problem of Veselago lenses in hexagonal altermagnets, a recently discovered paradigm of magnetism separate from ferromagnetism and antiferromagnetism.
- Delve into problems related to materials exhibiting tilted Dirac cones.